

Release readiness & end-to-end QA

Live application assessment | 7 October 2026

smolcafe.in • Customer ordering • Cashier • Kitchen KDS • Barista • Admin

RECOMMENDATION	HOLD RELEASE
Primary reason	Order edits persist quantities without matching totals, and mixed-station actions incorrectly advance other stations.
Evidence	Two labeled live QA orders, table create/edit, inventory threshold write, five role interfaces, screenshots, DOM observations, local source inspection and 21 simulation tests.
Scope caveat	This is a substantial exploratory test pass, not exhaustive certification. Physical phones, independent devices, load, offline behavior and payment-provider transactions remain unverified.

What matters most

Fix money and fulfillment correctness before visual polish. The application visibly propagates some changes, but several propagated values and statuses are wrong. Passing TypeScript and simulated business-rule tests did not catch these live failures.

Test-data notice

QA table: QA1007. Completed complimentary test order: #556. Cancellation was attempted for draft test order #753; it was still pending on the final readable check, and a retry could not be verified after browser automation stalled. Neither order represents money actually received. See the cleanup ledger before using revenue reports.

Prepared from direct UI actions and repository evidence. Local commit inspected: 1fd0fea. The deployed commit was not verified. Times below are observation bounds, not benchmark percentiles.

Executive findings

Priority	Finding	Why it blocks stabilization
P0	F01: Cashier edits do not update saved total	₹286 in editor becomes ₹203 after save, while quantity increases persist.
P0	F02: Station statuses are coupled	Kitchen start advances drinks; barista ready/delivered advances food.
P1	F03-F06: Payment, dispatch, notes and tax defects	Unpaid confirmation claims money received; food-only dispatch reaches Barista; notes absent; tax appears after submit.
P1	F07-F10: Guest, dietary and reporting defects	Guest has no active session; Veg filter retains egg; unpaid orders inflate revenue; live analytics contains static values.
P1	F11-F12: Customer editing / grocery persistence	Edit Order opens ordinary empty-cart menu; saved threshold reopens at old value.
P2	F13-F20: UI, data and responsiveness	Duplicate table cards, slow saves, dead search, planning notes, incorrect receipts and table recovery.

Positive behavior observed

All four staff roles accepted the supplied PIN. Named customer onboarding eventually retained the correct table. Menu category routing worked for Coffee First. Cart modifier and quantity arithmetic worked before submission. A mixed order reached Cashier and retained the modifier. Customer tracking received preparation and served updates. Table creation and seating edits persisted after reload. Barista timer start/pause/reset and station searches worked.

Release gate

Require successful retests of F01-F09 and F11-F12, using independent customer, cashier, kitchen and barista sessions. Do not accept screenshots of optimistic success as proof: verify persisted totals, item-level statuses, notes, payment state and audit records after reload.

Method, environment & evidence

Area	Executed
Live surfaces	https://www.smolcafe.in/ ; supplied customer link /smol-cafe/table-xyzabc; created QA link /smol-cafe/table-qa1007; role dashboards reached through UI.
Browser	Chrome desktop, with responsive CSS viewports. Confirmed customer / kitchen / admin mobile screenshots at 390 × 844 and Barista at 360 × 800. Desktop baseline approximately 1458 × 748.
Isolation	Separate role tabs in one browser profile. Cookies may be shared; this is not independent-device verification. Admin required re-login after another role login and reload.
Data	Used QA1007 DO NOT PREPARE, a reserved fictional US number, a dedicated QA table, and complimentary fulfillment. Existing orders #163 and #821 were inspected only.
Evidence	Screenshots and text snapshots under output/pdf/evidence. Source references in findings are local corroboration, not proof of deployed implementation.
Limitations	No load generation, destructive bulk actions, real payment, refund, vendor submission, push permission, real printer, physical camera scan, iOS/Android device or offline simulation.

Result language

Confirmed = directly observed in the live UI. Corroborated = live observation supported by local code. Suspected = observed discrepancy whose deeper cause remains unproven. Not run = a test plan, not a claimed result. P0 means release-blocking financial or fulfillment correctness; P1 means high-impact functional failure; P2 means medium-impact usability, consistency or reliability.

Timing caution

The browser control layer sometimes timed out even when navigation completed. Reported application waits use visible pending/success observations. They include UI/network/tool overhead and sparse sampling. No p50/p95/p99, server latency or true cross-device latency is claimed.

The live order journey

Step	Observed result
1. Establish QA table	Created QA1007 / smol-cafe / 2 seats; edited to 4. Duplicate in-memory cards appeared, but reload showed one saved four-seat table.
2. Named onboarding	QA1007 DO NOT PREPARE reached Home with correct table identity. Guest skip earlier failed to establish a session.
3. Build mixed basket	Espresso ₹79 + Bun Makkhan ₹79 + Extra Cheese ₹35 = displayed Grand Total ₹193.
4. Submit order #556	Customer confirmation claimed ₹203 received. Cashier showed Awaiting Approval; no money had been received. Correct modifier arrived.
5. Cashier edit	Espresso quantity 1 → 2; added QA instruction. Editor: ₹272 subtotal + ₹14 displayed tax = ₹286. Saved cards: updated quantities but old ₹203 total. Reload did not fix.
6. Dispatch food only	Selected Kitchen (Food), then Complimentary. Cashier removed the entire order from approval queue. Both Kitchen food and Barista espresso appeared.
7. Kitchen starts	Clicked START PREPARING only in Kitchen. Barista moved to BREWING without barista action.
8. Barista finishes	Clicked MARK BREW READY, then MARK DELIVERED. Food became ready and customer eventually showed SERVED as well.
9. Second order #753	Plain bun displayed ₹79 before submit, ₹83 after. Customer Edit Order opened ordinary menu without existing basket/editor. Cancellation reason was submitted; final outcome unverified.

This trace proves propagation exists, but does not prove correctness. It reproduces wrong totals, misleading payment confirmation and shared station progression with two realistic mixed-order roles.

Findings & reproduction

F01 P0 | Cashier item edits leave persisted total stale

Confirmed; survives customer reload

Setup: Order #556 at QA1007. Originally espresso ×1 + cheese bun ×1, total ₹203.

Reproduce: Cashier → Edit Order → increase espresso to 2 → enter preparation note → Save & Update Order → inspect cashier and customer, then reload customer.

Actual: Editor shows ₹286. Saved cashier/customer show espresso ×2 (₹158) and bun ₹114 but total remains ₹203. Customer header also retains “2 items.”

Expected: All persisted order, bill and UI totals must be recalculated atomically from the same edited lines, with an explicit tax/rounding rule.

Fix direction: Use a transactional server command with revision check; fail and roll back on any update error. In local app/cashier/actions.ts:266, order-update errors are logged, then line deletion/insertion proceeds; instructions argument is not saved. Check actual schema columns and all error returns.

Evidence: cashier-edit-result.txt; cashier-edit-stale-customer.txt

F02 P0 | One station advances another station's fulfillment

Confirmed in mixed order #556

Setup: QA mixed order has food in Kitchen and espresso in Barista.

Reproduce: Start food only in Kitchen; inspect Barista. Mark drinks ready in Barista; inspect Kitchen. Mark drinks delivered; inspect customer.

Actual: Kitchen START PREPARING makes espresso BREWING. Barista READY makes food ORDER READY. Delivery propagates to entire order.

Expected: Kitchen and barista line groups must progress independently; overall order ready only when all required groups are ready.

Fix direction: Persist item/station statuses separately; derive aggregate state from all line groups. Enforce transitions and optimistic revision checks on the server. Emit station/item identifiers, not only an order-wide status.

Evidence: kitchen-start-changes-barista.txt; barista-ready-changes-kitchen.txt

Findings & reproduction

F03 P1 | Pay-at-counter submit claims payment received

Confirmed UI; no payment made

Setup: Checkout for QA order #556 uses Send Order to Cashier.

Reproduce: Submit pay-at-counter order and watch the transient success modal before automatic Orders navigation.

Actual: Modal says PAYMENT VERIFIED & SETTLED, “₹203 Received!”, Paid via Cashier Desk, and a CSH transaction ID. Tracking and Cashier say payment/approval pending.

Expected: Submission should say “Order sent; pay at cashier.” Receipt and payment success require real settlement evidence.

Fix direction: Separate order acceptance, dispatch and payment events. Do not invoke payment celebration for a DRAFT order; never synthesize a settlement receipt from submission.

Evidence: Direct live observation at ~5.6 seconds after submit; qa-order-draft.txt corroborates pending state. Transient modal screenshot was not reliably retained.

F04 P1 | Kitchen-only dispatch also sends drinks

Confirmed

Setup: Order #556 has two espressos and one food item.

Reproduce: Cashier → Kitchen (Food) → Complimentary → Confirm & Dispatch → inspect both stations with QA1007 filter.

Actual: Food reaches Kitchen and espresso reaches Barista. Entire order disappears from cashier pending queue.

Expected: Only food lines should dispatch; unsent drinks remain actionable until explicitly dispatched.

Fix direction: Make dispatch per line/station with independent dispatched_at and actor fields. Retain pending remainder in cashier. Validate server target parameter; render only dispatched lines in station queries.

Evidence: food-only-dispatch-kitchen.txt; food-only-dispatch-barista.txt

Findings & reproduction

F05 P1 | Cashier preparation notes missing at both stations

Confirmed for saved QA instruction

Setup: Cashier edit form for #556.

Reproduce: Enter “QA1007 TEST ONLY - DO NOT PREPARE. Espresso no sugar. Bun extra cheese.” → Save → dispatch → inspect both tickets.

Actual: Neither Kitchen nor Barista ticket displays the note. Food modifier becomes “(Extra)” instead of “Extra Cheese.”

Expected: Persist and prominently show relevant preparation instructions and exact modifier names.

Fix direction: Save instructions in the same transaction as lines. Model modifiers as IDs plus immutable labels; do not infer them from truncated item names. Local editCashierOrderAction accepts instructions but omits it from update payload.

Evidence: food-only-dispatch-kitchen.txt; food-only-dispatch-barista.txt; local app/cashier/actions.ts:266

F06 P1 | Checkout grand total excludes tax until submission

Confirmed twice

Setup: Named QA session, regular priced items.

Reproduce: Build espresso + cheese bun; inspect View Bill, submit. Repeat with plain bun.

Actual: “Grand Total” ₹193 becomes ₹203. Plain bun ₹79 becomes ₹83. No tax line appears in pre-submit bill.

Expected: Show the exact payable amount, tax breakdown and rounding before committing.

Fix direction: Use one server quote shared by cart, submission and cashier. Pin quote version/expiry, revalidate and ask customer to accept any changed amount. Keep all arithmetic in paise.

Evidence: qa-order-before.txt; qa-order-draft.txt; second order #753 observation

Findings & reproduction

F07 P1 | Guest skip loses table session and shows false labels

Confirmed in initial guest flow

Setup: Open supplied /smol-cafe/table-xyzabc link.

Reproduce: Choose Skip & order as guest → Home → Menu → add item → View Cart → Orders.

Actual: Home says Table 01; cart says Table 07; Orders says No Active Table Session. Named onboarding later correctly retains XyzAbc/QA1007.

Expected: Anonymous guest must still get a valid table session and one consistent table label.

Fix direction: Create/resolve dining session before either guest branch navigates. Remove fallback table IDs from production UI; block checkout if session unresolved. Local Home fallback “01”, CartDrawer fallback “07”.

Evidence: guest-no-session.txt; checkout.txt; local app/home/page.tsx:14 and components/cart/CartDrawer.tsx:44

F08 P1 | Veg-only filter retains an egg omelette

Confirmed

Setup: Customer Menu with all categories.

Reproduce: Enable Filter Veg only → inspect Slow Mornings → open Creamy Cheese Mushroom Omelette.

Actual: Other egg dishes disappear but this omelette remains. Detail explicitly says Contains Egg and describes egg omelette.

Expected: All egg-containing dishes must be excluded by the vegetarian filter.

Fix direction: Use a canonical dietary enum/ingredient-derived flag shared by filters and detail badges; validate menu imports and aliases. Add table-driven tests for every egg dish and combo.

Evidence: veg-filter.txt; egg-detail.txt; egg-detail-mobile.png

Findings & reproduction

F09 P1 | Revenue/payment ledger treats unpaid orders as verified

Confirmed; source corroborated

Setup: Before QA, cashier has #163 / #821 awaiting approval.

Reproduce: Compare Cashier with Admin Payments; submit unpaid #556 and inspect Overview before settlement.

Actual: Admin marks pending orders VERIFIED UPI. Gross Revenue “Paid & Verified” rises from ₹14,719 to ₹14,922 on unpaid #556. Complimentary #556 later also adds ₹203 to cashier paid total.

Expected: Keep sales booked, cash collected, comps, refunds and outstanding balances separate; verified rows need actual transaction records.

Fix direction: Build ledger from payment events, not order status. Local app/admin/actions.ts:286 emits entries for all non-cancelled/rejected orders, manufactures TXN IDs and marks non-served orders VERIFIED.

Evidence: payments.txt; cashier-baseline.txt; unpaid-counted-revenue.txt

F10 P1 | Analytics presents static values as live metrics

Confirmed; source corroborated

Setup: Admin Overview / Analytics with one original table.

Reproduce: Compare Table Management and Analytics; read table utilization and KPI values.

Actual: Management: one table / eight seats. Analytics: 1/12 active with T01-T12. Reorder 34.2%, turn duration 38 min and +8.4% comparison are literals in local UI.

Expected: Every live metric must use actual records and declared date/timezone, or be explicitly unavailable.

Fix direction: Remove demo fallbacks. Query true table IDs and session durations; derive comparisons from actual prior periods. Use the same revenue definition across charts/cards.

Evidence: analytics.txt; local AdminTowerDashboard.tsx:2351,2355,2367,2649

Findings & reproduction

F11 P1 | Customer Edit Order link does not load an editor

Confirmed navigation failure; update submit not attempted

Setup: Draft order #753, one bun.

Reproduce: Customer Orders → Edit Order.

Actual: Navigates to /smol-menu?editOrder=8b2fc06b-d896-4ef8-992b-958b6c07f578 but shows ordinary menu, no existing basket, edit banner or Update Order control.

Expected: Load pending order into a clearly identified edit flow; save should update that order rather than create a new one.

Fix direction: Parse and authorize editOrder, fetch current revision, hydrate editor, and enforce DRAFT-only edits. Local smol-menu page accepts table/t but does not process editOrder.

Evidence: Direct live DOM and URL observation; local app/smol-menu/page.tsx

F12 P1 | Inventory threshold save does not persist visible value

Confirmed symptom; root cause not established

Setup: Single-Origin Arabica Espresso Beans safety minimum 5 kg.

Reproduce: Configure Safety Thresholds → set 5.1 → Save → wait for modal closure → Refresh inventory → reopen editor.

Actual: Card and reopened editor still show 5, not 5.1. Operation closed within the 16-second observation window; no lasting update was visible.

Expected: Successful save must persist precise decimals and read back the same value; failures should remain actionable.

Fix direction: Verify affected-row count, ingredient IDs, numeric schema and data source. Distinguish real inventory from seeded display fallbacks. Return saved record and surface server validation errors.

Evidence: inventory-threshold-not-persisted.txt

Findings & reproduction

F13 P2 | Table create/edit duplicates stale cards until reload

Confirmed; reload recovers

Setup: Create QA1007 then edit capacity 2 → 4.

Reproduce: Save new table once → close QR preview → search QA1007 → edit first card capacity → save.

Actual: Two QA1007 cards appear despite total table count 2 including XyzAbc. After edit, one says 2 seats/Available and the other 4/Occupied. Reload shows one 4-seat QA table.

Expected: Exactly one card per persistent table ID, updated consistently.

Fix direction: Deduplicate optimistic insert plus realtime reconciliation by ID. Replace state from authoritative response or upsert into keyed map; derive all summary counts from same state.

Evidence: qa-table-duplicate-cards.txt; qa-table-edit-stale.txt; qa-table-duplicate-cards.png

F14 P2 | Long write waits and insufficient progress feedback

Confirmed sampled waits

Setup: Table create/edit and customer onboarding on live site.

Reproduce: Submit once and observe disabled Adding/Saving/Entering states before success.

Actual: Create pending >25.2 s, observed success by 39.3 s. Edit pending >17.4 s, success by 41.1 s. Named onboarding pending >14.7 s, success by 33.1 s.

Expected: Prompt acknowledgment, bounded server timeouts and honest retry-safe progress.

Fix direction: Instrument action/RPC spans, query count and timeouts. Remove serial lookups where safe, add single-flight refreshes, and expose confirmed last-sync time. Benchmark independently before attributing cause.

Evidence: Recorded timestamped UI observations; timing table in this report

Findings & reproduction

F15 P2 | Menu Search button has no usable search input

Confirmed; source corroborated

Setup: Customer /smol-menu.

Reproduce: Tap Search menu and inspect page/keyboard/input controls.

Actual: No search field or overlay opens and content remains unchanged.

Expected: Open/focus an actual labeled search input and show filtered results / clear / no-results states.

Fix direction: Local handler focuses menu-search-input if present. Render the input or implement a real search panel; test mouse, keyboard and screen-reader activation.

Evidence: Direct live observation; local MenuClientView.tsx:206

F16 P2 | Internal planning notes exposed to customers

Confirmed

Setup: Customer menu cards across categories.

Reproduce: Open menu and inspect omelette, chai, oats and experimental drink cards.

Actual: Copy includes egg legality/permission to be confirmed, vendor cost targets, inventory advice and commercially justified matcha notes.

Expected: Customer copy should explain ingredients, taste and relevant dietary facts only.

Fix direction: Separate public description from staff/recipe/cost notes in schema and serializers. Review imported content before publish.

Evidence: veg-filter.txt; egg-detail-mobile.png and menu observations

Findings & reproduction

F17 P2 | Receipt identifier loses a character

Confirmed on existing order #181

Setup: Cashier Paid Orders.

Reproduce: Open Chit for ORD-181.

Actual: Receipt prints "Order ID: #RD-181" while table shows ORD-181.

Expected: Receipt and order identifiers must match exactly for reconciliation.

Fix direction: Use a canonical order display ID formatter; avoid arbitrary string slicing. Verify receipt/customer/admin identifiers together.

Evidence: Direct receipt DOM observation; cashier-paid.txt contains originating order

F18 P2 | Recovery menu offers nonexistent tables and wrong home

Confirmed for Table 01

Setup: Guest Orders shows no active session.

Reproduce: Click suggested Table 01. Inspect error and Back to Home destination from Orders.

Actual: Table 01 produces "Unable to start a dining session." Orders offers T01-T13 despite configured XyzAbc; Back to Home links to staff root /.

Expected: Offer only active tables or rescan assistance, and route customer home to /home.

Fix direction: Render recovery from actual table configuration; eliminate hardcoded IDs and keep customer/staff navigation distinct.

Evidence: guest-no-session.txt; invalid-recovery.txt

Findings & reproduction

F19 P2 | Catalog counts and upsell content disagree

Confirmed discrepancy; discount intent needs confirmation

Setup: Admin Menu and customer menu/cart.

Reproduce: Compare admin navigation badge to loaded catalog; compare Conversation Board menu and cart upsell.

Actual: Admin badge says 59, catalog/customer say 65. Menu Board is ₹329 with chickpea/vegetable description; upsell is ₹260, struck ₹350, cheese/fruits/nuts description.

Expected: One catalog source; explicit promotion eligibility and matching product description.

Fix direction: Use catalog SKU and promotion IDs rather than hardcoded upsell objects; update badge from loaded catalog. Validate allergens and recipe of promoted item.

Evidence: admin-menu.txt; checkout.txt; local CartDrawer.tsx:207-210

F20 P2 | Completed tickets retain ever-growing preparation timers

Confirmed

Setup: Kitchen completed list, 49 historical tickets before QA.

Reproduce: Open Kitchen and inspect completed historical orders.

Actual: Completed tickets show “17354m timer” and similar multi-day elapsed values. Large empty columns consume most mobile screen.

Expected: Freeze prep duration at readiness/completion; archive old tickets and show useful timestamps.

Fix direction: Compute prep SLA from explicit started/ready timestamps, render age separately, and collapse empty mobile lanes/history by default.

Evidence: kitchen-baseline.txt; kitchen-mobile.png

Measured waits & synchronization

Action	Visible observation	Interpretation
Create table	Pending at 25.25 s; success by 39.27 s	Long operation; one click only.
Edit table seating	Pending at 17.39 s; success by 41.07 s	Persisted after reload; UI duplicate stale.
Named onboarding, XyzAbc	Pending at 14.74 s; Home by 33.12 s	Table correct once complete.
QA-table onboarding	Home observed by 24.00 s	Upper observation bound only.
Order #556 submit	Payment modal visible by 5.65 s	Misleading success copy; order is draft.
Cashier arrival	Absent at ~5.65 s; present by 20.62 s	Shared browser tabs; not cross-device SLA.
Cashier edit	Saving at 5.93 s; closed by 24.30 s	Quantities propagated, total did not.
Kitchen-only dispatch	Kitchen visible by 9.37 s	Barista also received it once filter changed.
Kitchen start → Barista	BREWING observed by 7.21 s	Fast enough to expose wrong coupling.
Barista ready → Kitchen	ORDER READY observed by 7.25 s	Wrong station-level transition.
Inventory save	Modal closed by 16.09 s	Value still old after refresh/reopen.

Targets to agree and measure next

Proposed targets, not measured guarantees: input feedback under 100 ms; order submit p95 under 2 s on normal café connectivity; foreground cross-device convergence p95 under 2 s / p99 under 5 s; no silent stale state beyond 10 s. Test at least 100 representative operations per class and record failures, not just successful samples.

Local polling fallbacks: Cashier 8 s, Kitchen 6 s, Barista 4 s, customer Orders 15 s, plus realtime subscriptions. A “Live 3s” cashier badge therefore does not alone establish a three-second freshness guarantee. Audit subscription health and refresh concurrency.

Executed test register

ID	Scenario	Result / evidence
C01	Supplied QR landing	PASS: XyzAbc page opens.
C02	Guest skip and table identity	FAIL F07/F18: no session; fallback labels/recovery.
C03	Named international test onboarding	PASS functional; slow F14.
C04	Coffee First category link	PASS: Coffee, Obviously selected.
C05	Search menu	FAIL F15: no search input opens.
C06	Veg-only filter	FAIL F08: egg omelette remains.
C07	Item detail/modifier price	PASS: ₹79 + ₹35 = ₹114.
C08	Cart increase/decrease/remove	PASS: ₹114 → ₹228; removal shows empty cart.
C09	Checkout totals / submit	FAIL F03/F06: undisclosed tax and false paid modal.
C10	Customer pending-order edit	FAIL F11: no order editor loaded.
C11	Track preparation/served	PARTIAL: updates arrive, but incorrect aggregation F02.
S01	Cashier/Kitchen/Barista/Admin PIN	PASS login for all four. No brute-force testing.
S02	New mixed order in Cashier	PASS: #556 and exact modifier/quantities arrive.
S03	Cashier quantity/notes edit	FAIL F01/F05.
S04	Food-only dispatch	FAIL F04: also sends drink.
S05	Independent Kitchen/Barista state	FAIL F02 across start/ready/delivered.
S06	Cashier paid list / receipt	PARTIAL: renders; totals/reporting/ID issues.
S07	Barista timer start/pause/reset	PASS: running, paused 4.7 s, reset 0.0 s.
S08	Kitchen and Barista searches	PASS: QA filters and nonmatch empty state.
S09	Kitchen catalog / cookbook	PASS opens: 65 dishes; cookbook four example recipes.
S10	Cancel QA #753	ATTEMPTED: reason accepted by tool; final persistence not verified.

Executed test register

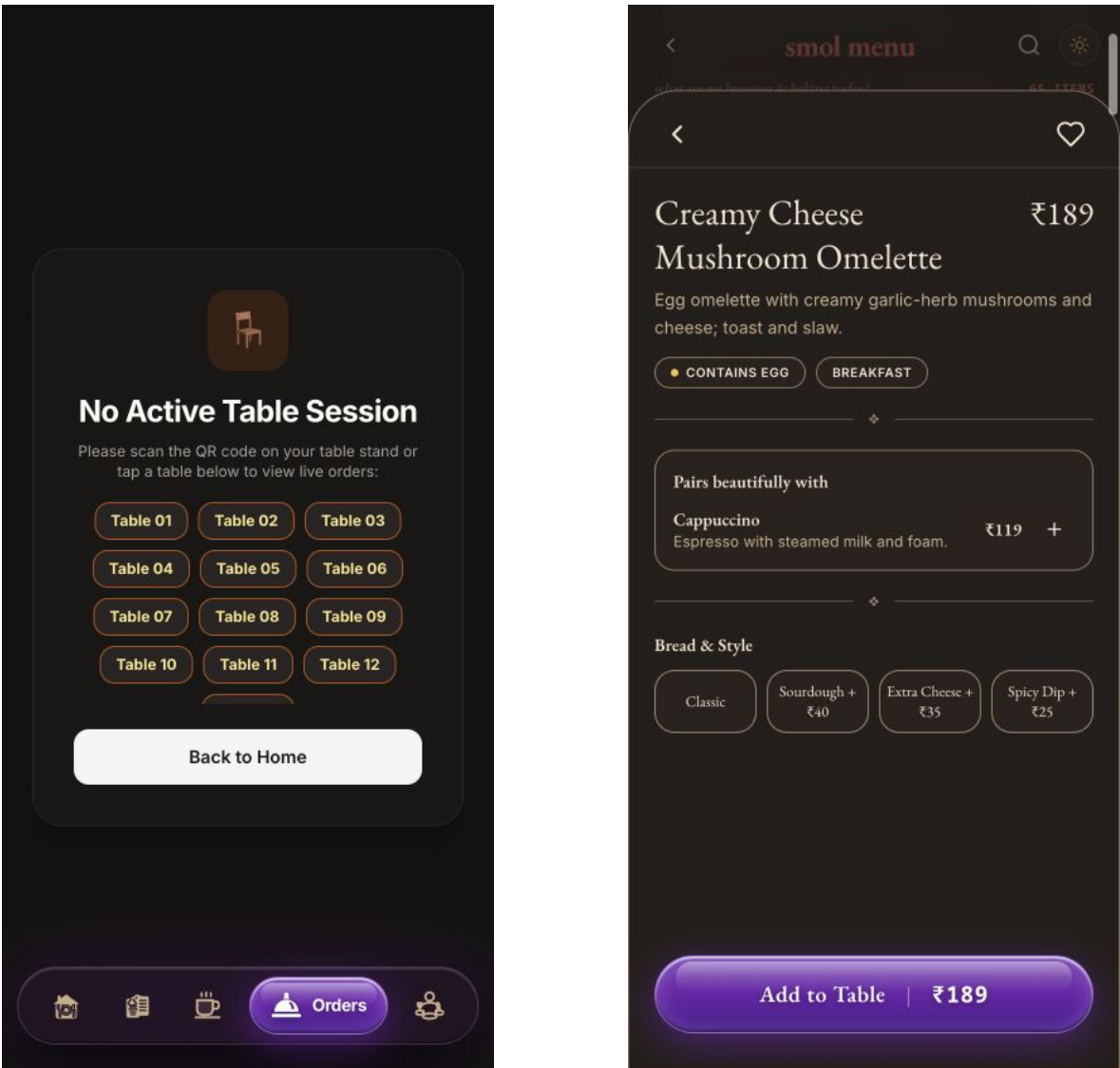
ID	Scenario	Result / evidence
A01	Table list / Add Table form	PASS after load; transient empty state initially.
A02	Create QA1007 / generated QR	PASS persisted; slow and duplicate cards F13/F14.
A03	Edit seats 2 → 4	PASS after reload; stale duplicate before reload.
A04	Generated QA customer URL	PASS: correct table and named onboarding. Physical QR scan not tested.
A05	Table search	PASS: QA1007 isolated.
A06	Grocery radar / threshold form	PASS opens; write does not visibly persist F12.
A07	Refresh/reopen threshold	FAIL F12: value remains 5 instead of 5.1.
A08	New purchase-order form	INSPECTED: vendor, lines, quantity, cost, date. Cancelled unsaved; no vendor commitment.
A09	Admin catalog counts	FAIL F19: 59 badge / 65 loaded.
A10	Analytics / utilization	FAIL F10: static metrics and phantom table grid.
A11	Payments / revenue	FAIL F09: pending orders called VERIFIED.
A12	Mobile navigation / inventory	INSPECTED at 390 px; dense labels, overflowed nav tabs, fixed bottom bar.
T01	Invariant simulation	PASS 7/7: dedupe, price change, stock race, status transition, webhook, history.
T02	Refund simulation	PASS 2/2: partial and full refund / loyalty reversal.
T03	RBAC simulation	PASS 8/8: negative role matrix cases.
T04	Webhook simulation	PASS 4/4: HMAC, capture, duplicate and ordering.
T05	TypeScript	PASS all workspaces via npm run typecheck.

Why simulation passes did not mean release ready

The invariant suite builds an in-memory SmolSimulationDatabase. These scripts validate their own modeled rules, not the deployed Next.js actions, real SQL transactions, station subscriptions or rendered workflows. Add integration tests against the actual schema and browser tests against a seeded staging deployment.

Commands run: npm test; npm run test:refunds; npm run test:rbac; npm run test:webhook --workspace=@smol-cafe/db; npm run typecheck. All exited successfully. No source fixes or deployment were performed.

Mobile evidence



Guest journey

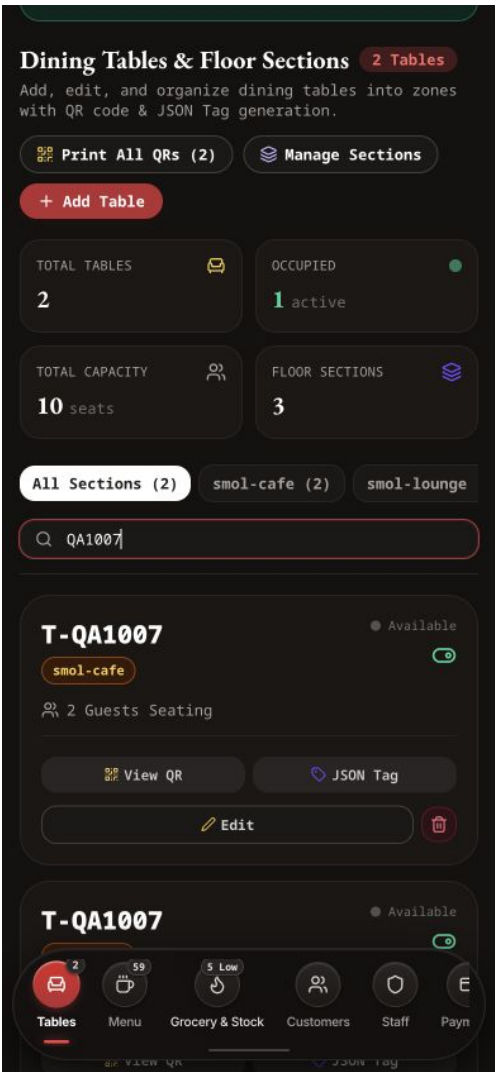
Guest skip reached menu but Orders had no active table session. Suggested hardcoded table recovery was invalid.

Dietary consistency

Veg-only search retained an omelette. Its detail explicitly states Contains Egg, proving filter/detail disagreement.

Screenshots are CSS viewport checks, not physical-device or mobile-Safari certification.

Admin & kitchen evidence



Duplicate table rendering

One creation generated duplicate QA1007 cards in the current view. A reload removed the duplicate; later seating edit had stale and current cards simultaneously.



Kitchen vertical space

Empty New and Preparing sections consume most of the 390 px screen. Staff must scroll far to see later lanes; use compact empty states and sticky counts.

Mobile-first improvements

Surface	Change to implement	Acceptance check
Customer	Persistent real table chip; scan/session validation before checkout. Remove false default table numbers.	Every screen, invoice and station uses the same table ID/label.
Checkout	One sticky payable amount with subtotal, tax, discount and rounding. Separate “Send order” from “Pay.”	Pre-submit and persisted total match to the paise.
Order tracking	Per-item or per-station progress; clear awaiting approval/payment copy. Keep feedback/reviews below essential status.	Food preparing + drink ready shown accurately; no full-ready notification early.
Menu	Working search, dietary chips, public copy only, exact modifier labels and reliable promotion descriptions.	Filter and detail classifications agree for every dish.
Cashier	Mobile order cards with visible edit/approve actions; sticky save summary; changed-total warning.	No important action hidden offscreen or behind a bottom bar.
Kitchen/Barista	Compact lane tabs with counts, large primary action, persistent notes, separate age/prep timer.	One-handed use at 360 px; important notes visible without opening another screen.
Admin	Short bottom-nav labels, clear “More” menu, horizontal-scroll cues and mobile forms with sticky footer.	Reach tables, groceries and analytics at 360/390 px without accidental navigation.
All roles	Loading skeletons, retry-safe pending state, error details, confirmed sync timestamp and reconnect badge.	Network loss cannot show unconfirmed success or silently stale data.

Accessibility acceptance checklist

Target at least 44 px primary touch areas; visible keyboard focus; labeled icon buttons; dialog titles, focus trap and Escape close; announcement of save/error/order updates; reduced-motion splash; adequate contrast; text scaling at 200%; no fixed overlay covering focused fields or footer actions. These are proposed acceptance checks, not a completed accessibility audit.

Regression suite to build

Group	Test steps	Required invariant
Concurrency	Two independent cashiers edit same revision; one approves while customer edits.	Conflict is explicit. No lost update, duplicate lines or partial total.
Idempotency	Double tap, retry after response timeout, reload during pending submit; replay same key.	Exactly one order and one financial event; safe status recovery.
Station routing	Food only, drink only, mixed, combos, partial dispatch, station reassignment.	Correct destinations; unsent remainder actionable; independent status.
Order editing	Add/remove lines; modifiers; decimals; zero/negative/large qty; price change mid-edit.	Server validates all fields; totals/bill/stock/history atomically match.
Payment	Cash, UPI sandbox, card sandbox, partial payment, complimentary, refund, split bills.	Financial state independent of order state; no fabricated payment ID.
Session isolation	Two patrons same table; two tables same browser; fresh device; expired/disabled/renamed QR.	No cross-table leakage; correct active session and bill.
Offline/reconnect	Drop network before submit, after commit, during edit, during status update.	Retry dedupe; no false success; eventual reconciliation.
Background tabs	Switch apps, lock phone, reconnect Wi-Fi, resume after 30 min.	Foreground refresh reconciles missed events and exposes stale state.
Inventory	Concurrent last-stock orders; recipe depletion; modifier ingredients; void/restock/spoilage.	No oversell; deterministic units; stock ledger matches reservations.
Procurement	Draft PO, partial receipt, overreceipt, duplicate receipt, vendor filtering, costing.	No stock change until receipt; no duplicate accounting; audit trail.

Run destructive, load and payment-provider tests against staging with disposable fixtures. The live site is not a safe substitute for repeatable test isolation.

Device, analytics & security matrix

Area	Required cases before signoff
Mobile devices	iPhone Safari 360/390/430-class widths; Android Chrome 360/412; portrait/landscape; software keyboard; safe-area inset; installed PWA; font scaling.
Admin desktop	1366 × 768 and 1440+; tab order; modal bounds; zoom 125/200%; long tables; printer preview.
Analytics	IST midnight boundary, backdated orders, date filters, no data, cancelled orders, unpaid drafts, comps/refunds, historical price snapshots; reconcile each metric to raw ledger.
Authorization	Fresh anonymous/customer/kitchen/barista/cashier/admin contexts. Direct server-action denial, stale session, logout in another tab, ownership checks, CSRF assumptions and audit attribution.
Table lifecycle	Create duplicate label, rename occupied table, zone move, disable active table, duplicate QR scan, close/reopen session; QR download/printed camera scan.
Menu lifecycle	Price/availability change propagates to customer and KDS; deleted item already in cart; long names; missing images; modifier stock; dietary import validation.
Receipts	Order IDs, timestamps/timezone, CGST/SGST and rounding, discounts/comps, refunds, split tender, download/print on actual hardware.
Scale / endurance	Agreed peak concurrent patrons plus staff; 8-hour shift; websocket disconnect storm; polling overlap; memory growth; long histories.
Notifications	Opt-in/denial, browser background, sound muted, duplicate ready events, all stations not yet ready, cancelled order.
Data recovery	Database transaction failure halfway through edit; failed stock RPC; retry after deploy; migration compatibility; backup restore rehearsal.

Security note tied to observed scope

All roles use the supplied shared PIN in this test setup. Before production signoff, verify distinct staff identities/roles and audit attribution. No brute-force, credential rotation, penetration test or API bypass was performed. A role login in one tab affected an admin reload, so independent browser contexts are required for trustworthy authorization testing.

Engineering stabilization plan

1. Financial integrity first: repair cashier edit transaction and schema mapping; save notes; recompute order/bill totals together; never proceed after an update error. Add integration tests that reproduce #556 exactly.
2. Correct state model: split payment_status, order_status, dispatch state and item/station status. Require all station groups before aggregate READY/SERVED. Test concurrent changes with revision/version checks.
3. Honest customer contract: resolve guest session before navigation, show server-quoted payable total, remove payment celebration from pending orders, implement real pending-order editor.
4. Unify source data: canonical dietary classification, real table list, shared catalog/promotion data, actual payment ledger and analytics definitions. Remove production demo fallbacks.
5. Repair realtime reconciliation: commit first, emit versioned event, invalidate relevant queries, upsert by ID, discard older responses and reconcile on foreground/reconnect. Same-browser broadcast is only a supplement to cross-device delivery.
6. Instrument measured slowness: trace each server action, SQL/RPC latency, retries, payload size and refresh overlap. Use one in-flight refresh per surface, debounce correctly, batch reads, and bound timeouts.
7. Add confidence layers: unit tests for pure totals/state reducers, real database integration tests, independent-context UI regression, then physical mobile and realistic load tests.

Local code leads worth investigating

app/cashier/actions.ts:266: nonatomic edit and omitted instructions. app/admin/actions.ts:286: synthesized payment ledger. components/admin/AdminTowerDashboard.tsx:2351 onward: static KPI/table values. components/cart/CartDrawer.tsx:44,207: table/upsell fallbacks. components/menu/MenuClientView.tsx:206: search target. lib/queries/inventory.ts:26: stock RPC errors return 9999, a fail-open risk requiring integration verification.

These paths are relative to apps/web in local commit 1fd0fea. They identify likely fix locations; confirm deployed code and database schema before changing production.

Cleanup ledger & handover

Artifact / change	Last verified state	Required follow-up
QA table QA1007	Created in smol-cafe; seating edited to 4; occupied after onboarding.	Close QA dining session and disable/archive table through normal admin workflow after review. Not deleted.
Customer QA1007 DO NOT PREPARE	Named onboarding completed with reserved fictional test number.	Treat as QA customer; exclude from marketing and customer analytics.
Order #556	SERVED; Complimentary; 2 espressos + 1 cheese bun; saved total incorrectly ₹203.	Preserve as bug evidence; exclude from revenue/stock analysis. No actual cash/UPI/card payment occurred.
Order #753	Draft plain bun ₹83; still pending on final readable check. Cancellation retried; result unverified after browser timeout.	Check Cashier/Admin now; cancel if still pending. Do not prepare.
Espresso bean threshold	Attempted 5 → 5.1; refresh/reopened editor still 5.	Verify authoritative value remains 5; no intended permanent change.
Existing #163 / #821	Read-only inspection of pending orders.	No QA changes intended or made to these orders.
PO / restock / stock quantity	New PO form inspected and cancelled; no PO, vendor send, stock receipt or physical stock adjustment submitted.	No purchasing cleanup needed.
Code / deployment	No application source edits, migrations, credential changes or deployment.	Only report/evidence output files were created.
Browser state	Several role/customer test tabs; final cancellation control timed out.	Temporary responsive viewport override was reset successfully. Close remaining QA tabs; cancellation verification still requires attention.

Signoff status

NOT READY FOR RELEASE. Money and fulfillment defects are reproducible in live workflows. This report does not claim all use cases passed or that all user-requested workflows were fully exercised. Use the executed register as evidence and the remaining matrix as the next acceptance plan.

Suggested next cycle: implement F01-F09/F11-F12 → seed a clean staging café → replay mixed-order tests in four independent device contexts → verify financial/stock ledgers → run physical mobile and latency/load checks → issue a retest report.